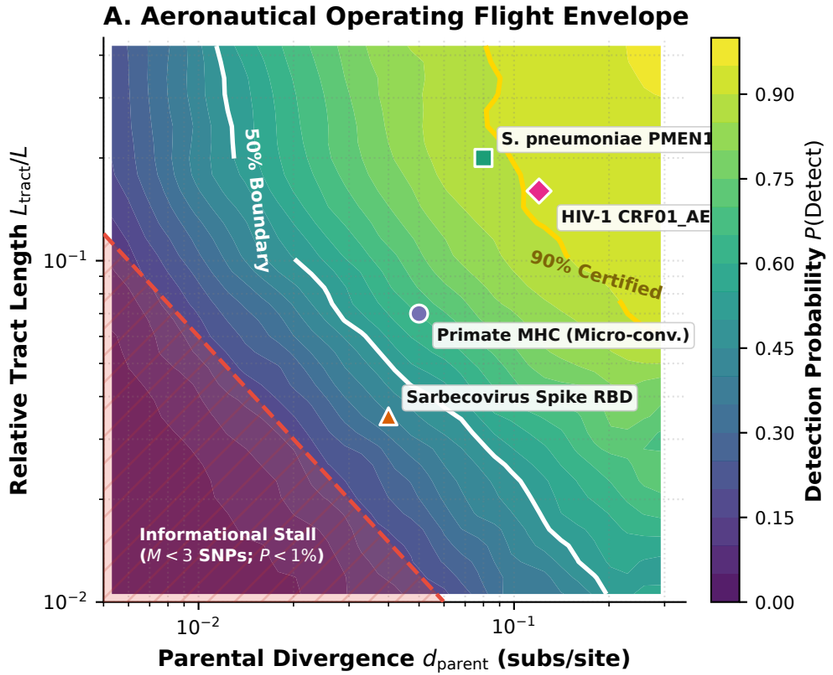
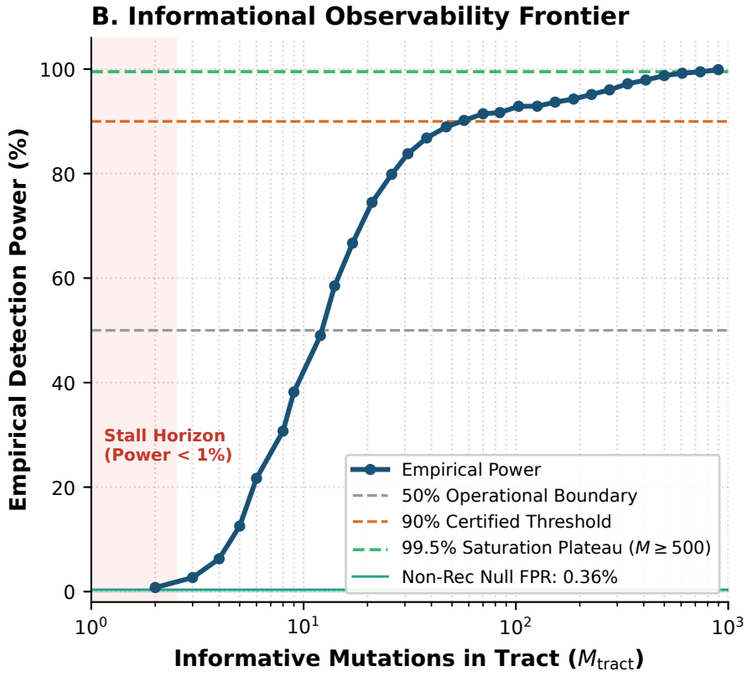
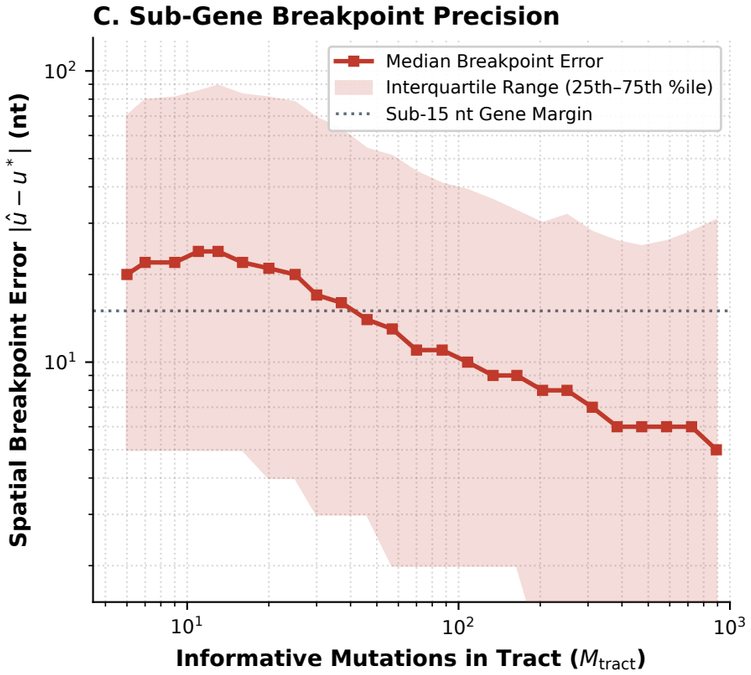

A. Aeronautical Operating Flight Envelope
S. pneumoniae PMEN1
HIV-1 CRF01_AE
Primate MHC (Micro-conv.)
Sarbecovirus Spike RBD
50% Boundary
90% Certified
Informational Stall
($M < 3$ SNPs; $P < 1\%$)
Relative Tract Length L_{tract}/L
Parental Divergence d_{parent} (subs/site)
Detection Probability P(Detect)


B. Informational Observability Frontier
Stall Horizon
(Power < 1%)
Empirical Power
50% Operational Boundary
90% Certified Threshold
99.5% Saturation Plateau ($M \geq 500$)
Non-Rec Null FPR: 0.36%
Empirical Detection Power (%)
Informative Mutations in Tract (M_{tract})


C. Sub-Gene Breakpoint Precision
Median Breakpoint Error
Interquartile Range (25th–75th %ile)
Sub-15 nt Gene Margin
Spatial Breakpoint Error $|\hat{u} - u^*|$ (nt)
Informative Mutations in Tract (M_{tract})